

- 1. Outline.
 - I Introduction and Background
 - II Governing equations for partial melting problem
 - i Flow
 - ii Transport
 - iii Phase
 - III Numerics
 - i Flow (BR and Uzawa solver)
 - ii Transport (MLWENO)
 - iii Eutectic phase package
 - iv Coupled computation
 - IV Computer implementation details
 - i PETSc
 - ii Test problems
 - iii parallelization
 - V Application to real problems
 - i Mid-ocean ridge problem
 - VI Summary, conclusion and future work

2. Finite Volume with MLWENO reconstruction. A general scalar advection-diffusion equation can be represented as follows:

$$(2.1) \quad u_t + \nabla \cdot (F(u) - D(u)\nabla u) = 0, \mathbf{x} \in \mathbb{R}^d$$

which can be discretized with finite volume scheme.

$$(2.2) \quad \bar{u}_{E,t} + \frac{1}{|E|} \int_{\partial E} (\hat{f}(u) - \nabla u \cdot \nu_E) d\sigma(\mathbf{x}) = 0$$

where piece-wise constant solution $\bar{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}, t) d\mathbf{x}$. We choose Lax-Friedrichs scheme to represent our numerical flux scheme. We approximate integrals over each facet with quadrature rule.

$$\bar{u}_{E,t} + \sum_{j=1}^{L_E} \frac{e_j}{|E|} \sum_k \omega_{j,k} [\hat{f}(\mathcal{R}^+, \mathcal{R}^-) - \mathcal{D}(u)] = 0$$

where

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